

A District-Level Analysis of Child Malnutrition in India

Problem Statement and Data exploration

India presents a major development paradox: despite rapid economic growth over the past two decades, child malnutrition remains widespread, with high rates of stunting and anaemia. This persistence, often called the "**South Asian Enigma**," has been linked to multiple structural factors, including poor sanitation, low female empowerment, food insecurity, and weak healthcare systems.

However, most existing studies focus on single factors or use national- or state-level data. What remains lacking is **district-level evidence** that simultaneously evaluates the roles of **food availability, water and sanitation, and healthcare access** in shaping nutrition outcomes.

This gap has important policy implications. Large government programs such as the **Jal Jeevan Mission** and **POSHAN 2.0** are investing heavily in water access, nutrition, and healthcare. Without district-level evidence identifying the primary constraint in each region, resources risk being allocated inefficiently.

This project therefore asks: **How do differences in agricultural productivity, rural water access, and healthcare coverage explain variations in child malnutrition across Indian districts, and which factor acts as the key constraint?**

Conceptual Framework

Three primary pathways are hypothesized to explain district level variation in child malnutrition outcomes.

- **Pathway 1 - Food availability**

Agricultural productivity influences food availability and nutrition in households. Districts with persistently low crop yields may experience structural food insecurity regardless of market access. This Pathway is proxied using district level weighted crop yield from agricultural statistics dataset.

- **Pathway 2 - Water and Sanitation**

Access to safe drinking water and improved sanitation reduces exposure to enteric diseases that may impair nutrient absorption and contribute to stunting, even when food intake is adequate.

- **Pathway 3 - Healthcare reach**

Healthcare system coverage also influences nutrition through maternal and child health services such as antenatal care, immunisation and micronutrient supplementation.

Women’s education is also included as a structural socioeconomic control rather than a separate causal pathway as it likely influences all the pathways simultaneously as educated mothers are more likely to adopt improved health, nutrition and sanitation practices.

Datasets and Sources

Dataset	Source	Geography	Period	Role in Analysis
NFHS-5 District Factsheets	NDAP	District (704)	2019–21	Primary outcome variables
NFHS-4 District Factsheets	NDAP	District (637)	2015–16	Temporal change analysis
Agriculture Statistics	NDAP	District	2019	Food availability pathway
HMIS 2019–20	NDAP	District	2019	Healthcare reach pathway
Jal Jeevan Mission Dashboard	NDAP	District	2025 snapshot	Water access pathway

All five datasets were merged at the district level using normalised state-district name keys. Match rates: NFHS-5 100%, NFHS-4 99.7%, HMIS 99.1%, Agriculture 96.2%, JJM 91.2%.

Exploratory Data Analysis

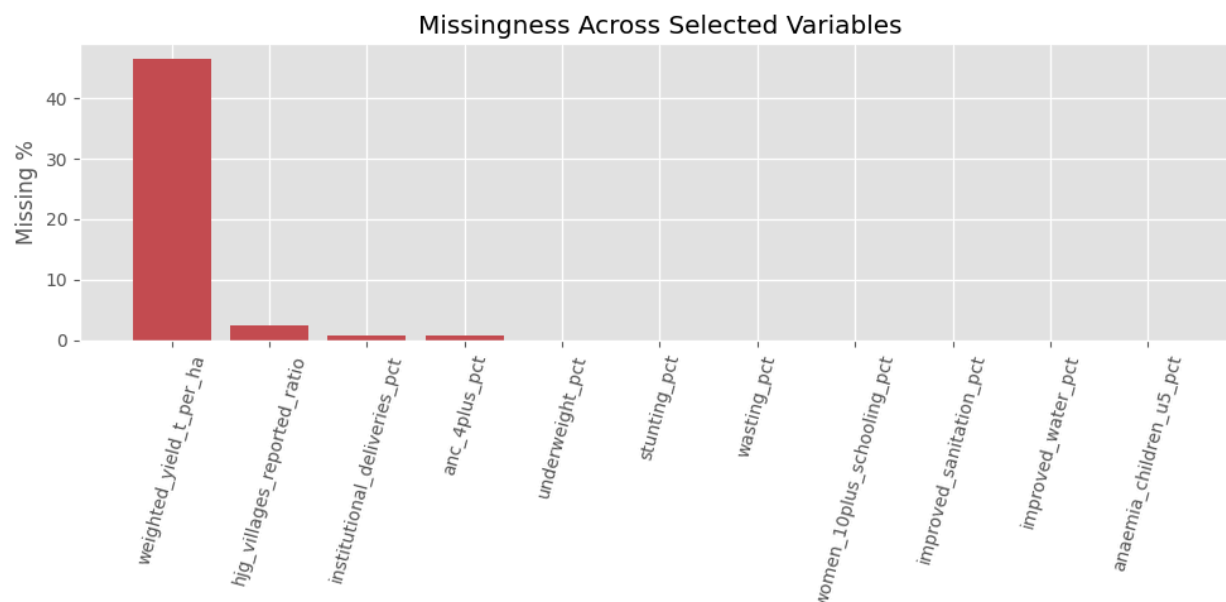
1. Data Coverage and Merge Quality

Source	Total Rows	Matched	Unique Districts Matched	Match Rate
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NFHS-5	704	704	704	100.00%
NFHS-4	637	635	635	99.70%
HMIS 2019	704	698	698	99.10%
Agriculture 2019	10,797	10,565	376	96.2% (unique districts)
JJM 2025	753	687	687	91.20%

The merged data have 704 districts across 36 states and union territories. We used NHFS-5 as the reference geography. Match rates were high across the datasets which exceeded 99% for NHFS-4 and HMIS, while JJM matched about 91% of districts but was mainly due to administrative boundary changes in andhra pradesh.

The main coverage limitation arose from agricultural data, which was available only for 376 unique districts which reduced the final regression sample to 370 districts. Most other variables such as nutrition, sanitation and socioeconomic indicators have near complete coverage due to which descriptive analysis can be conducted on full sample but models including agricultural yield are restricted to the smaller subset.



The missingness profile across key analysis variables

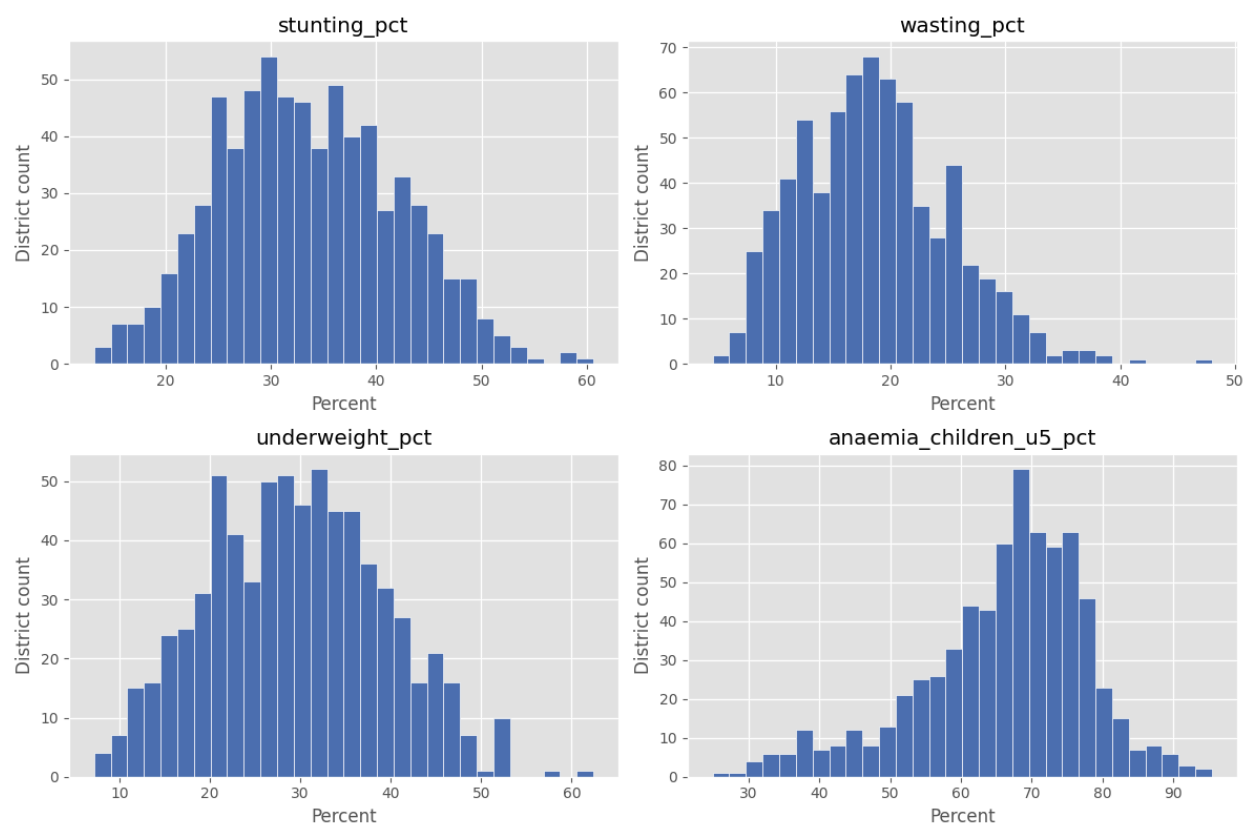
2. Current Burden of child malnutrition

Indicator	Mean	SD	Min	Median	Max
Stunting (%)	33.5	8.5	13.2	32.8	60.6
Wasting (%)	18.6	6.5	4.5	18.1	48

Underweight (%)	29.6	9.7	7.2	29.5	62.4
Child anaemia (%)	66	12	24.9	67.7	95.5
Improved sanitation (%)	71.9	14.3	29.2	73.7	99.9
Women 10+ years schooling (%)	40.2	14.2	13.6	39	88.2
Institutional deliveries, HMIS (%)	93.6	10.6	27.9	98.4	100

District level nutrition indicators show that child malnutrition remains widespread and highly unequal across India where stunting averages 33.5%, underweight 29.6%, wasting 18.6% and child anemia 65.9% with substantial variation between districts.

The histograms below show the full district-level distribution of all four indicators:

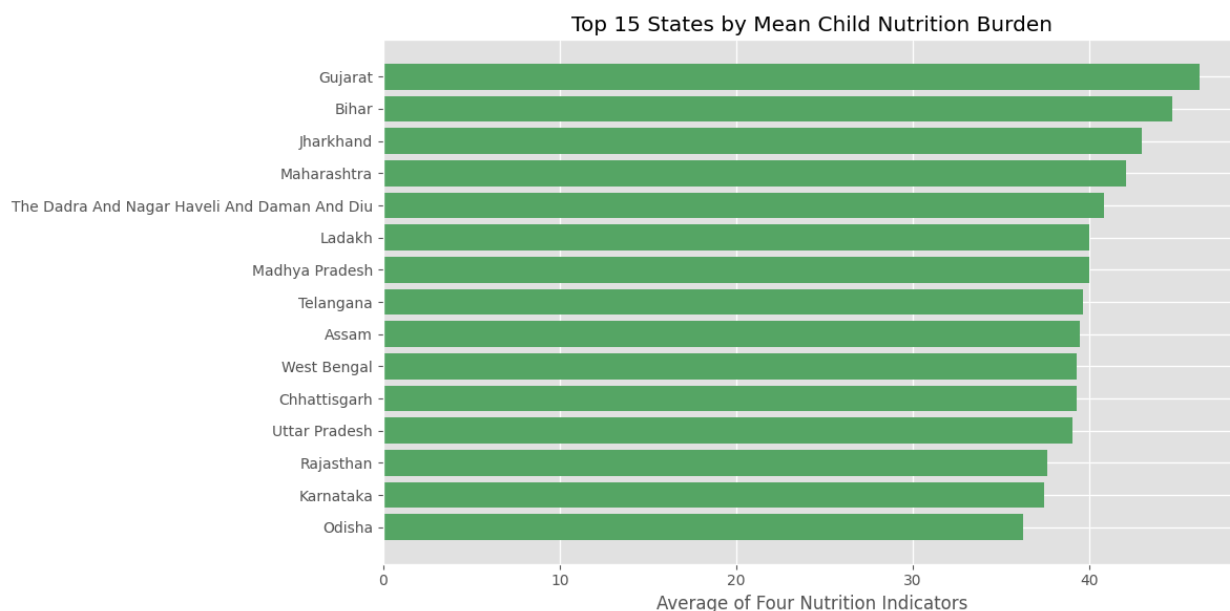


3. Temporal Trends NFHS-4 to NFHS-5

Indicator	Mean Change	Districts Improving
Stunting	-2.58 ppt	~65%
Wasting	-2.18 ppt	~58%
Underweight	-3.26 ppt	~68%
Child anaemia	9.35 ppt	~33%

Between NFHS-4 (2015-16) and NFHS-5 (2019-21), most districts showed moderate improvements in anthropometric indicators where stunting, wasting and underweight slightly declined on average however, child anaemia increase significantly, with only about one third of districts showing some improvement.

4. Geographic Variation



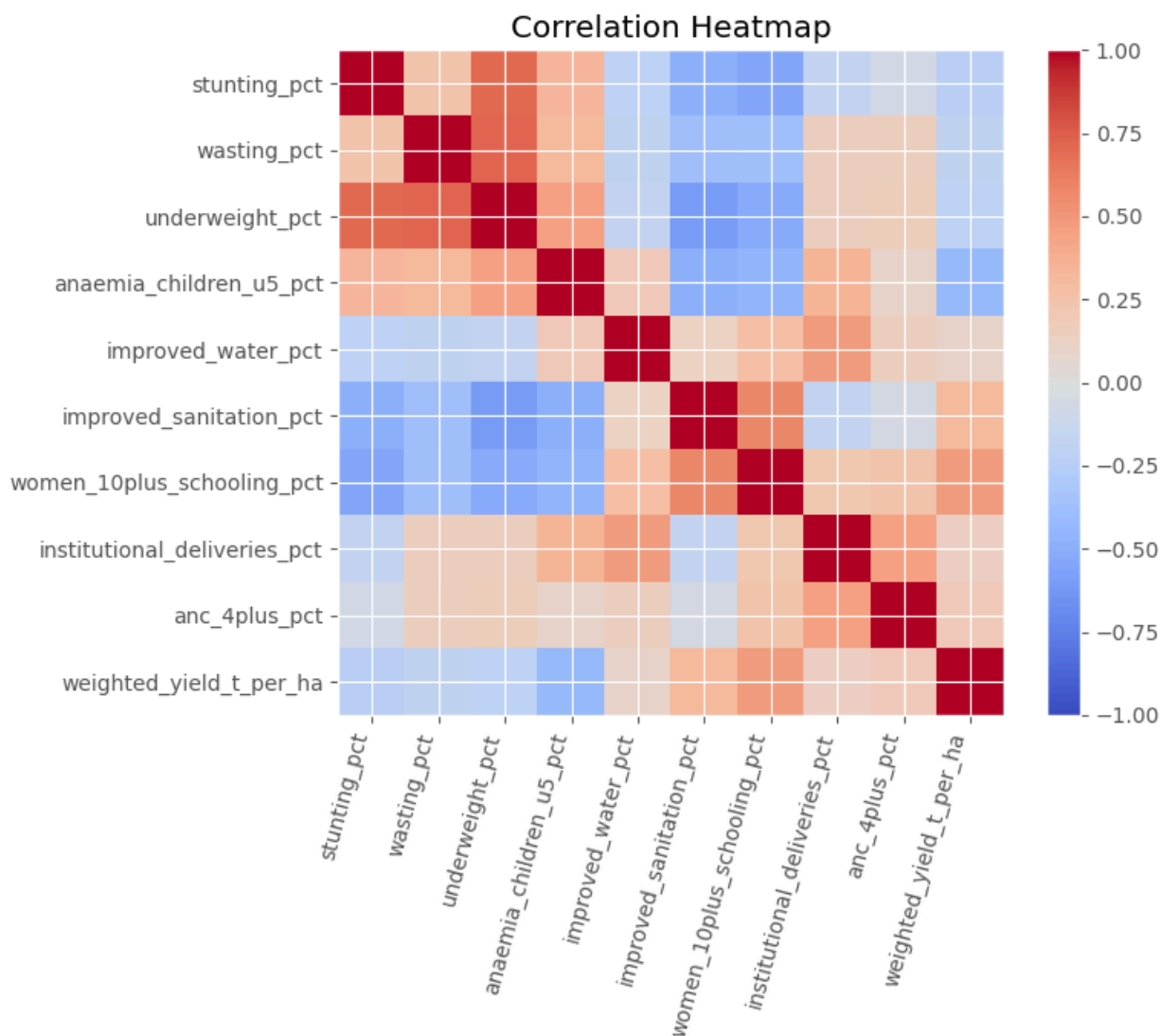
State level aggregation shows geographic differences in malnutrition burden. Figure above shows the fifteen highest burden states ranked by their average across the four nutrition indicators. High burden districts are concentrated in Jharkhand, Gujarat, Bihar, Maharashtra and Uttar Pradesh.

The highest burden districts which were ranked using the composite malnutrition index (PC1) are listed in table below. In contrast, districts which are part of kerala, Punjab, Haryana and parts of northeast show consistently lower malnutrition levels.

District	State	Malnutrition Index (PC1)
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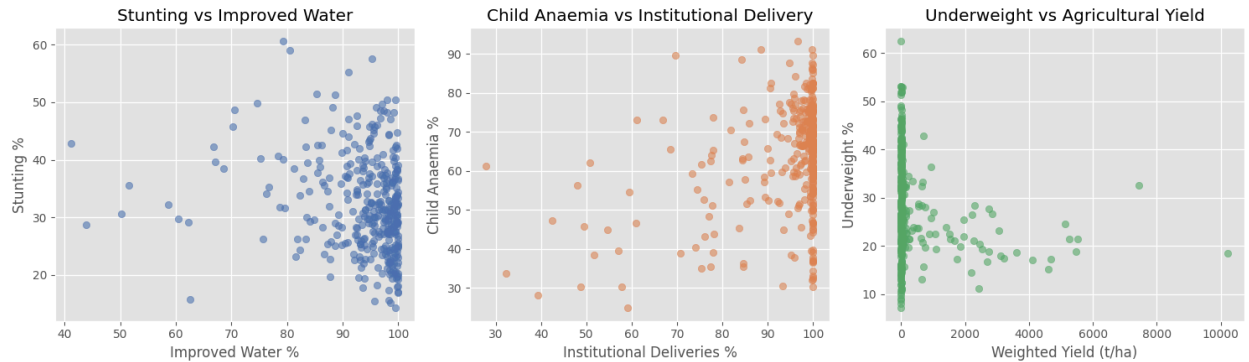
West Singhbhum	Jharkhand	4.74
Panch Mahals	Gujarat	4.28
Dohad	Gujarat	4.11
Nandurbar	Maharashtra	4
Tapi	Gujarat	3.69
Arwal	Bihar	3.51
Arvalli	Gujarat	3.44
Banda	Uttar Pradesh	3.34

5. Correlates of Malnutrition: Bivariate Patterns



multiple outcomes.

Among the explanatory variables, improved sanitation and women's education shows strongest negative relationships with malnutrition indicators. Water access shows weaker associations which is due to limited variation and agricultural yield shows mixed relationships.



The Scatter plots above show selected bivariate relationships. The stunting-water plot shows a weak negative trend, the anaemia-delivery plot shows little visible association and the underweight-yield plot shows a wide spread which suggests skewness and possible outliers in agricultural data.

Database Design and Analysis

Database Architecture

The Project is using sqlite database as the central data store. The database contains 12 tables which includes the raw source tables, analysis tables and metadata tables. The districts table serves as the geographic reference, with all other tables linked using district identifiers to ensure consistent joins. The analysis_district table stores merged values while analysis_district_normalized contains z-score standardised variables used for PCA, regression and clustering.

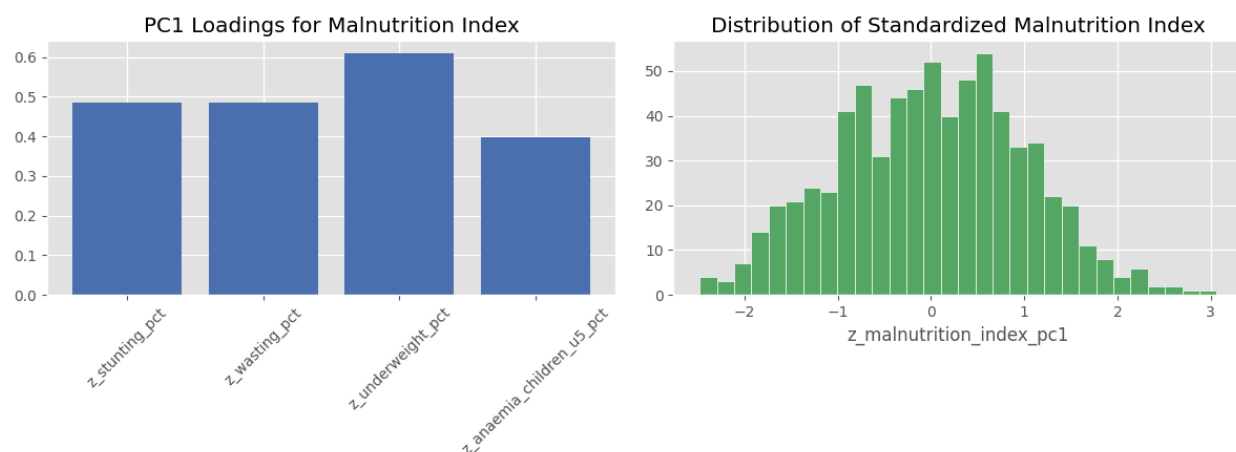
Table	Description	Rows
districts	Reference geography: 704 NFHS-5 districts with normalised keys	704
nfhs5	NFHS-5 district nutrition and household indicators	704

nfhs4_total	NFHS-4 district indicators (total residence only)	635
agriculture_2019	District-level crop production and yield data, 2019	10,565
hmis_2019	District healthcare indicators, HMIS 2019–20	698
jjm_2025	JJM village tap water coverage snapshot	687
analysis_district	Merged analysis-ready table with all raw values	704
analysis_district_normaliz ed	Same table with z-score standardised columns	704
nutrition_change_4_to_5	NFHS-4 to NFHS-5 change for 635 overlapping districts	635
normalization_summary	Mean, SD, and non-null count for each normalised variable	23
merge_summary	Source-level merge coverage summary	5
merge_unmatched	Unmatched district records by source, for auditing	—

District names across datasets were harmonized using automated cleaning and alias matching, achieving above 90% match rate across all sources. Any unmatched records were logged separately.

Composite Malnutrition Index via PCA

To simplify analysis, a composite malnutrition index was created using principal components analysis (PCA) applied to four standardised nutrition indicators - stunting, wasting, underweight and child anaemia.



Indicator	PC1 Loading
Underweight	0.610
Stunting	0.485
Wasting	0.484
Child anaemia	0.398

The figure above shows the PCA loadings and index of resulting index. The first principal component (PC1) explains 59.8% of total variance, with all indicators contributing positively. Underweight has the largest loading while anaemia contributes the least indicating partial differences in its trend as compared to other indicators.

Regression analysis: Identifying binding constraints.

1. Pathway specific Bivariate models

Model	Predictor	n	R ²	β	p-value
Sanitation pathway	Improved sanitation	704	0.356	-0.597	< 0.001
Education (control)	Women 10+ years schooling	704	0.321	-0.567	< 0.001

Agriculture pathway	Weighted crop yield	376	0.098	-0.321	< 0.001
Healthcare pathway	Institutional delivery rate	698	0.002	0.044	0.251

Separate regression models were estimated to examine each predictor independently and the results are summarised in the table above. Sanitation and women's education shows the strongest relationships with the malnutrition index where each explains over 30% variance. Agricultural yield shows a weaker but statistically significant relationship, while institutional delivery rates show almost no independent association.

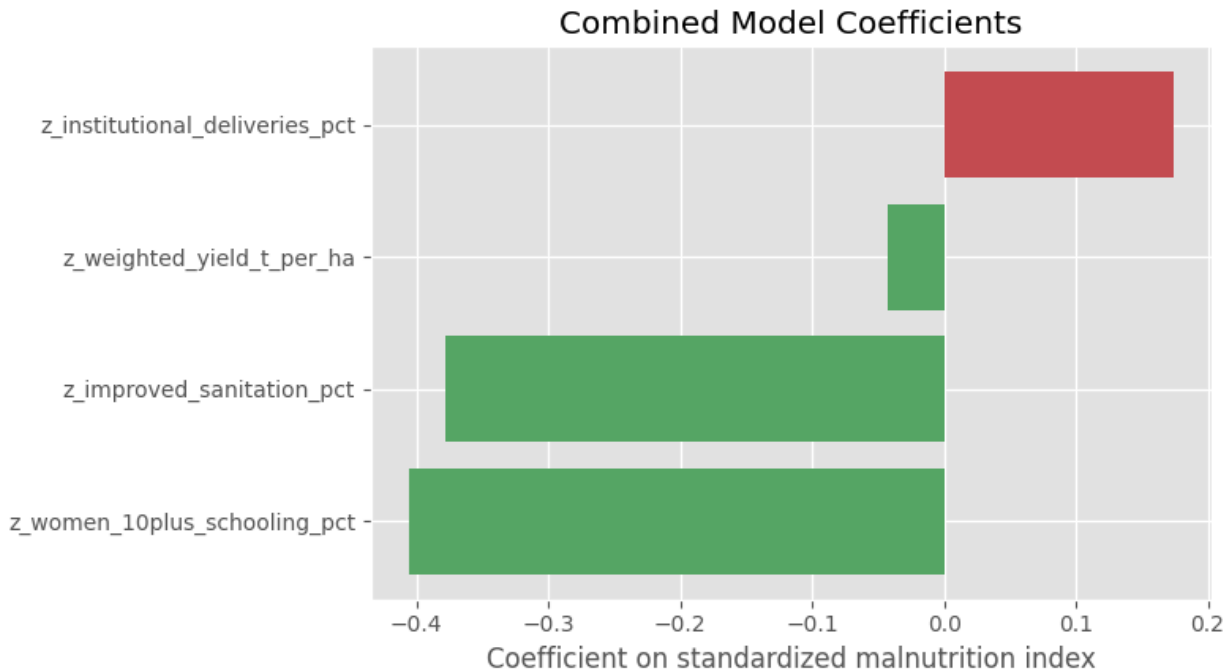
These results tell us that sanitation and education are the strongest single predictors when examined separately.

2. Combines cross-sectional model

All predictors were included simultaneously in a combined regression model using districts with complete data where (n=370). Model results are summarised in table below and coefficients, magnitudes, and directions can be visualized in figure below.

Variable	Coefficient	Std. Error	t-statistic	p-value
Intercept	-0.028	0.039	-0.731	0.466
Women 10+ years schooling	-0.406	0.049	-8.367	< 0.001
Improved sanitation	-0.378	0.052	-7.307	< 0.001
Institutional delivery rate	0	0	4.641	< 0.001
Agricultural yield	-0.043	0.043	-1.003	0.316

Model fit: $R^2 = 0.523$, Adjusted $R^2 = 0.518$, RMSE = 0.708, n = 370



The combined model is explaining approximately 52% of the variation in the malnutrition index ($R^2 \approx 0.52$), which indicates strong explanatory power when structural and pathway variables are considered together. Among the predictors, women's education and improved sanitation remain highly significant where both show large negative coefficients which suggests that district with higher female schooling levels and better sanitation coverage tend to have substantially lower malnutrition burden.

In contrast to this, agricultural yield shows only a weak effect after controlling for other factors and institutional deliveries rates display a positive coefficient which indicates a more complex relationship than it was initially expected.

Overall, the combined model highlights that structural conditions explain malnutrition more effectively than it was initially expected.

3. Interpreting the coefficients

The regression coefficients highlight the relative importance of different predictors in explaining malnutrition outcomes at district level.

A. Women's education is the strongest predictor, which means with higher level of female schooling strongly associated with lower malnutrition burden. This reflects improved health awareness, better nutritional practices and more effective use of available services.

B. Sanitation also remain independently significant, showing that districts with better

sanitation access are more likely to have lower malnutrition levels. This supports the role of sanitation in reducing disease exposure that affects nutrient absorption.

C. Agricultural Yield, although it is significant in bivariate analysis, it become insignificant in the combined model. This suggests that its earlier relationship with malnutrition may be due to its association with broader structural factors rather than direct effects.

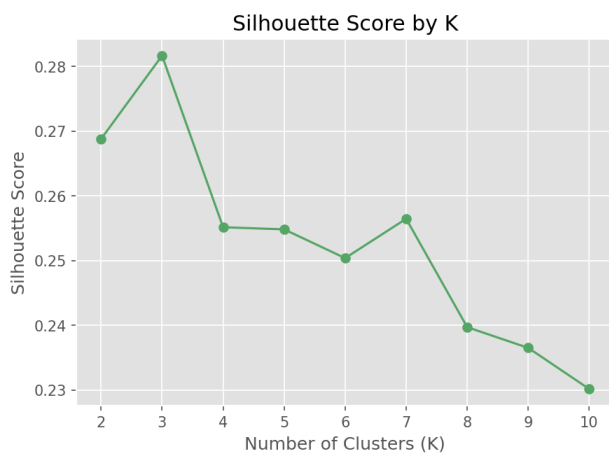
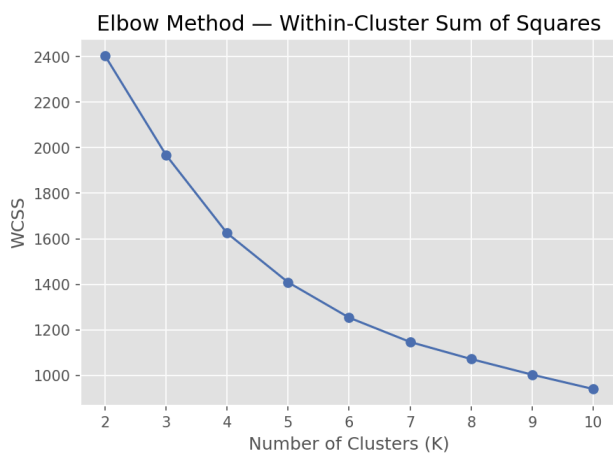
D. Institutional delivery rates show a positive coefficient which may reflect higher healthcare investment in historically high burden districts rather than a direct causal relationship.

Overall, these findings indicate that education and sanitation are the most consistently associated factors, while agricultural productivity and delivery rates show weaker or indirect relationships.

District Vulnerability Profiling via K-Means Clustering

K-means clustering was applied to 681 districts with complete data using five standardised variables:

1. The composite malnutrition index
2. Sanitation coverage
3. Women’s education
4. Institutional delivery rates
5. JJM tap water coverage



K	WCSS	Silhouette

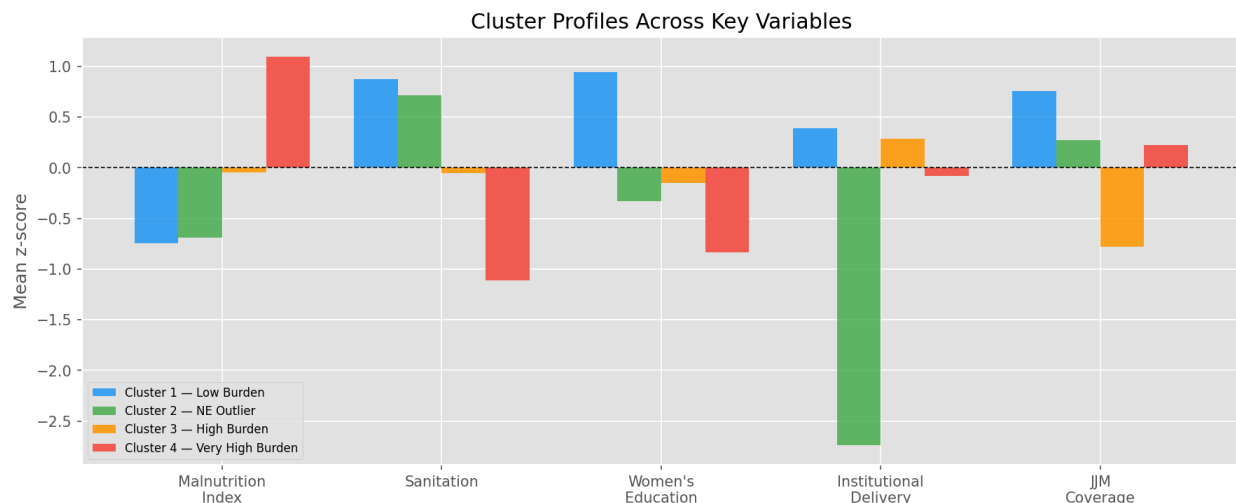
2	2,404.5	0.269
3	1,968.2	0.282
4	1,624.8	0.255
5	1,408.8	0.255

To determine the number clusters (**k**), both the elbow method and silhouette scores were evaluated for values from **K=2 to 10**. As shown in figure above, the silhouette score peaks at k=3 while elbow curve stabilises at k=4. Therefore, K=4 was selected as it provides clearer and more interpretable district groupings. The clustering metrics are summarised in table above, and the resulting cluster characteristics are shown in table and figure below.

Cluster profiles: Clusters are labelled 1–4 in ascending order of mean malnutrition burden.

Cluster	Label	n	Malnutrition Index	Sanitation	Women's Education	Delivery	JJM
1	Low Burden	196	-0.74	+0.88	+0.94	+0.39	+0.75
2	NE Outlier	50	-0.69	+0.71	-0.33	-2.74	+0.27
3	High Burden	253	-0.05	-0.06	-0.15	+0.28	-0.78
4	Very High Burden	182	+1.10	-1.11	-0.84	-0.08	+0.22

(All values are mean z-scores; positive = above national average, negative = below)



Four distinct profiles emerge which are:

1. Cluster 1 - Low Burden (196 districts)

This districts show high sanitation and high female education, along with strong water access and low malnutrition levels. They represent structurally advantaged regions and are commonly found in states such as kerala, Punjab, Haryana, Tamil Nadu, and Himachal Pradesh.

2. Cluster 2 - Northeast Outlier (50 districts)

This cluster shows relatively low malnutrition despite low institutional delivery rates and moderate education levels. These districts are primarily located in the northeastern states where cultural practices and dietary diversity may contribute to low malnutrition despite the weaker formal healthcare indicators.

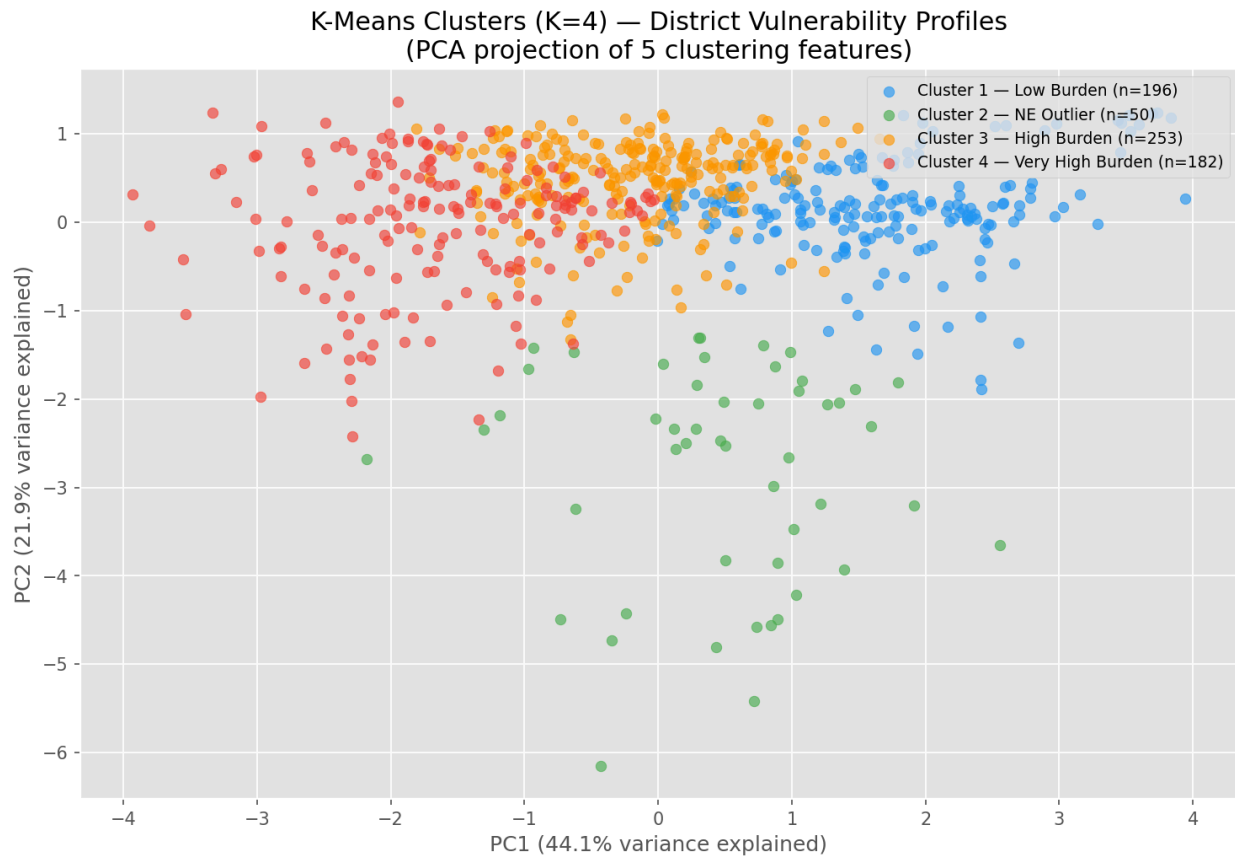
3. Cluster 3 - High Burden (253 districts)

This is the largest cluster with near average values across most variables but relatively lower water coverage. These districts represent moderately disadvantaged regions without strong structural advantages and are mostly distributed across central and eastern india.

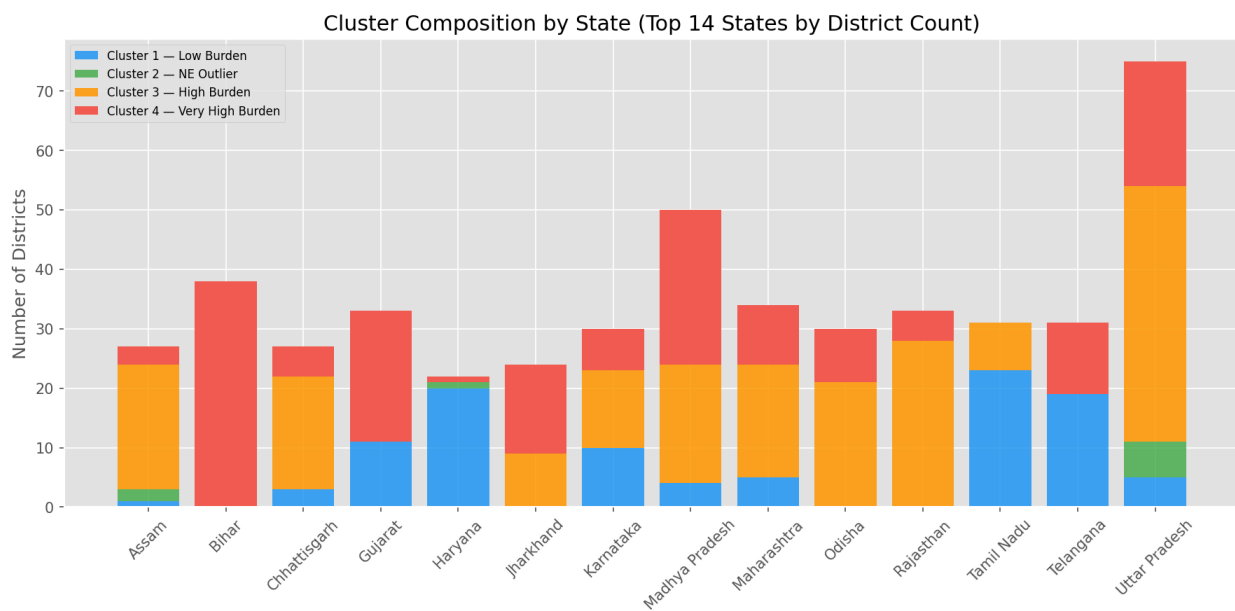
4. Cluster 4 - Very High Burden (182 districts)

These districts show very high malnutrition combines with low sanitation and low female education making it the most vulnerable group. This cluster represents structurally deprived regions where multiple disadvantages occur simultaneously.

The separation of these clusters in feature space is visualised in the plot below where low and high burden districts appear clearly separated along the main variation axis.



State level distribution of clusters is shown in figure below, revealing strong regional patterns. Here some states are dominated by higher risk clusters while others consist mainly of low burden districts. The highest burden districts are listed in table below highlighting areas with most severe outcomes.



Cluster	District	State	Malnutrition Index (z)
4 — Very High	West Singhbhum	Jharkhand	+3.07
4 — Very High	Panch Mahals	Gujarat	+2.77
4 — Very High	Dohad	Gujarat	+2.66
4 — Very High	Dang	Gujarat	+2.53
4 — Very High	Nandurbar	Maharashtra	+2.44
3 — High	Chandrapur	Maharashtra	+2.02
3 — High	Nashik	Maharashtra	+1.38
3 — High	Kurnool	Andhra Pradesh	+1.32

Recommendations and limitations

Insights from the analysis

The analysis shows us that child malnutrition across indian districts is strongly linked to structural and social conditions, particularly women's education and sanitation access. These factors consistently show stronger relationships with malnutrition outcomes than agricultural productivity or healthcare delivery indicators.

Another important finding is the large variation between the districts within same state which indicates that national or state level averages can mask severe local differences, making the district level analysis essential for understanding the true distribution of malnutrition burden.

A particularly notable result is the persistence of child anaemia, which shows weaker improvement as compared to other nutrition indicators. This tells us that micronutrient

deficiencies remain an important challenge even where overall growth related indicators show moderate progress.

Implications and recommendations

The findings suggest that child malnutrition should be understood as a multi factor problem which is influenced by sanitation, education, healthcare and food systems instead of any single variable.

Sanitation shows a strong association with lower malnutrition levels which suggests that improving access to sanitation facilities and encouraging consistent usage may contribute to better health outcomes. Coordination between water supply and sanitation services also may further strengthen these effects.

Women's education emerges as the most consistent predictor across models which tells us that improving female secondary education and literacy may have long term benefits for nutrition through better health awareness and household decision making.

Agriculture shows weaker independent effects in the combined model which suggests that improving dietary diversity and access to nutrient rich foods may be more relevant than focusing only on total crop output.

Healthcare services, particularly the ones which are related to micronutrient supplementation and child nutrition monitoring may require closer attention, especially given the persistent burden of anaemia observed in the data.

According to these results, it suggests that integrated improvements across sanitation, education and nutrition services is like to be more effective than isolated interventions.

Limitations of analysis

There are several limitations which should be considered when interpreting the results:

1. The analysis relies primarily on cross sectional data which is identifying statistical relationships but it does not establish direct causal effects. Some observed relationships may be influenced by unmeasured factors such as governance quality, geography, or historical developmental pattern
2. Data availability varies across sources, particularly for agricultural indicators which reduces our sample size for regression models and it may effect the generalisability of certain results.

3. The healthcare variables in the dataset may not fully capture the aspects of service quality that are most relevant to nutrition outcomes. Similarly the agricultural data used also does not reflect dietary composition or food diversity, which are likely important for addressing micronutrient deficiencies.